

Energy Valve™

Analysis Tool

User Guide · V10.0.20

for Belimo_Energy_Valve_Analysis_Tool.xlsm

Covers Energy Valve EV2, EV3, EV4 and EV4.2

Belimo Automation — for internal and partner use

Contents

Contents.....	2
1. About this guide.....	4
2. Before you begin.....	4
What you need	4
Enabling macros.....	4
Saving is switched off — on purpose.....	4
3. Quick start.....	4
4. The Dashboard	6
The panels, left to right.....	6
Choosing what the charts show	7
The button bar along the bottom	7
The small icons, top right.....	7
5. Importing data	8
Step 1 — choose the files	8
Step 2 — confirm the units	8
Step 3 — pick your time zone (first import only).....	8
Step 4 — wait.....	8
Step 5 — the Import Details form.....	9
Step 6 — the sensor mapping check.....	9
6. The Savings report	11
The measured values	11
The automatic diagnostics	12
Producing the deliverable.....	12
7. Data Integrity	13
8. Advanced Mode	13
Valve Analysis	14
Delta T Analysis	14
Delta T Threshold.....	14
Delta T Determination	15
Cost Calculation	15
Data Integrity.....	15
9. Dialog reference.....	15
10. Settings you can change.....	16
Energy cost inputs.....	16

11. Housekeeping	16
Clearing data	16
Versions	16
12. Troubleshooting	16

1. About this guide

This guide walks you through the Energy Valve Analysis Tool one screen at a time: what each sheet is for, what every dialog asks you for, and how to read the numbers it produces.

You do not need to understand Excel formulas or macros to use the tool. Everything happens through buttons and dialogs. Where the tool makes a judgement about your data — for example, whether the supply and return sensors are the right way round — this guide explains what triggered it and what to do about it.

Read sections 2 and 3 before your first import. After that, use the rest as a reference.

2. Before you begin

What you need

- **The tool itself** — Belimo_Energy_Valve_Analysis_Tool.xlsm. The file name stays the same from release to release; the version is shown at the bottom of the Dashboard. When a newer version is published, the tool tells you once when you open it and can install it for you; Check for Updates on the About screen does the same at any time.
- **Your valve's day files** — the CSV logs downloaded from the Energy Valve. You can select many at once — the tool merges them in date order.
- **Microsoft Excel for Windows** — with macros enabled. The tool is entirely macro-driven and will not work with macros blocked.

Enabling macros

If a yellow SECURITY WARNING bar appears across the top of Excel when you open the file, click Enable Content. If Excel blocks the file outright because it came from the internet or a network share, close it, right-click the file in Windows Explorer, choose Properties, tick Unblock at the bottom of the General tab, click OK, and re-open.

Saving is switched off — on purpose

You cannot save this workbook. If you press Ctrl+S you will get a message telling you so, and nothing will be written. This is deliberate: it guarantees that the copy in the shared folder is always a clean, empty template, so the next person never inherits somebody else's customer data.

The workflow the tool is built around is:

import → review on screen → print or e-mail the PDF → close without saving.

To analyse a second valve, just import again — the new data replaces the old.

3. Quick start

The whole job, end to end:

1. Open the workbook and click Enable Content if prompted.
2. Click IMPORT DATA (top-left of the Dashboard).
3. Select all the day files for the valve you are analysing, and click Open.
4. Confirm the units you want to see, then click OK.
5. The first time only: pick your time zone and click Ok.
6. Fill in the Import Details form — customer, equipment name and design figures — then click OK.

7. Review the Dashboard, then click SAVINGS for the customer-facing report.
8. Use the PDF or printer icon to produce the deliverable, then close the file without saving.

A typical import takes between fifteen seconds and a couple of minutes depending on how many days you selected. Progress messages appear in the status bar at the bottom of the Excel window and on the Dashboard itself.

4. The Dashboard

The Dashboard is the home screen. It is the only sheet visible when you open the tool, and every other screen is reached from it.

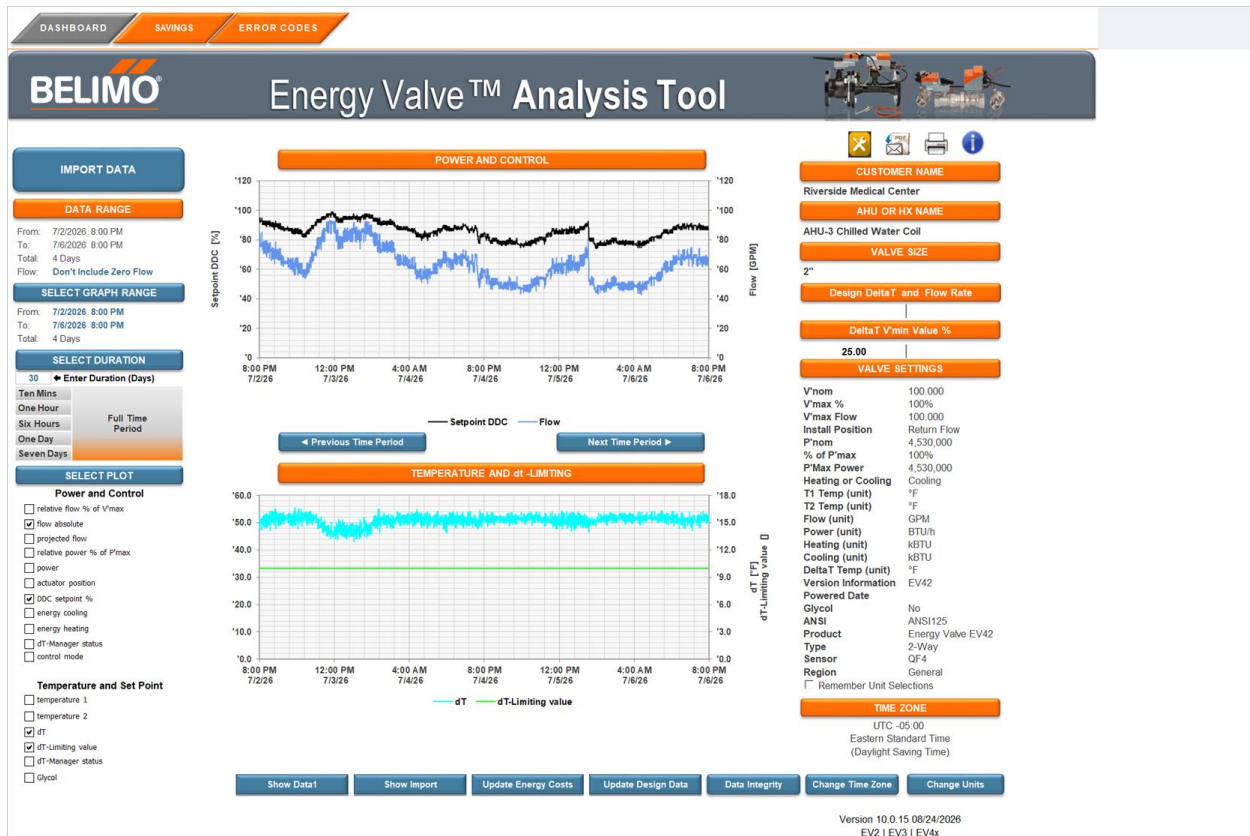


Figure 1 — the Dashboard after an import. Controls down the left, the two timeline charts in the middle, the valve's own settings down the right, and the action buttons along the bottom.

The panels, left to right

Panel	What it is for
IMPORT DATA	Starts an import. This is where every job begins.
DATA RANGE	The full span of the data you imported, and how many days it covers.
SELECT GRAPH RANGE	The window the charts are currently showing. Change the From and To dates here to zoom in on a period of interest.
SELECT DURATION	Preset zoom buttons — Ten Mins, One Hour, Six Hours, One Day, Seven Days, or Full Time Period. Enter Duration (Days) sets a custom span.
SELECT PLOT	Tick boxes choosing which traces appear on the two charts.
CUSTOMER NAME / AHU OR HX NAME	What you typed on the Import Details form. These appear on every report and in the PDF file name.
VALVE SIZE, Design DeltaT and Flow Rate, DeltaT V'min Value %	The design figures you entered. Correct them at any time with Update Design Data.
VALVE SETTINGS	Read straight off the valve: nominal and maximum flow, power limits, where the meter is installed, whether it is heating or cooling, the units it logs in, and its firmware generation.

Panel	What it is for
TIME ZONE	The zone the timestamps have been converted into.

Choosing what the charts show

The SELECT PLOT list drives both charts. Tick or untick and the charts redraw immediately. The traces available are:

relative flow % of V'max · flow absolute · projected flow · relative power % of P'max · power · actuator position · DDC setpoint % · energy cooling · energy heating · dT-Manager status · control mode · temperature 1 · temperature 2 · dT · dT-Limiting value · dT-Manager status · Glycol

Ticking everything at once makes the charts unreadable. Start with flow absolute and DDC setpoint % on the top chart, and dT with dT-Limiting value on the bottom one.

The button bar along the bottom

Button	What it does
Show Data1	Opens the imported data table — one row per logged sample. Useful for spot-checking a figure. No password needed.
Show Import	Opens the raw merged CSV exactly as it came off the valve, before any processing. Use it when you suspect a data problem.
Update Energy Costs	Change the electricity price, pump efficiency and pump head used for the dollar figures. See section 10.
Update Design Data	Re-opens the Import Details form so you can correct the customer name, design ΔT, design flow or design power without re-importing.
Data Integrity	Runs a quality check over the imported data. See section 7.
Change Time Zone	Re-selects the time zone. The timestamps are re-converted.
Change Units	Re-opens the unit chooser. Unit changes take effect on the next import, not retrospectively.

The small icons, top right

Icon	What it does
Spanner	Advanced Mode — reveals the engineering sheets. See section 8.
Envelope	Creates a PDF of the current sheet and attaches it to a new e-mail.
Printer	Prints the current sheet.
Blue i	About box — version and contact details.

5. Importing data

Step 1 — choose the files

Click IMPORT DATA. A file dialog opens. Select every day file belonging to the valve — click the first, then Shift-click the last — and click Open. The tool merges them and sorts everything by timestamp, so the order you select them in does not matter.

Import only files from one valve at a time. Mixing valves produces meaningless results.

Step 2 — confirm the units

The Unit Selection dialog appears. Choose how you want temperature, flow, power and energy displayed. Tick Remember Unit Selections if you want the same choices next time.

Field	Typical choice
Temperature	°F in North America, °C elsewhere
Flow	GPM in North America, m ³ /h or l/min elsewhere
Power	BTU/h or kW
Energy	kBTU or kWh
Remember Unit Selections	Tick to reuse these settings on the next import

Step 3 — pick your time zone (first import only)

Valve logs are always recorded in UTC. The tool asks once which zone to convert them into. Choose your zone from the list and tick Daylight Saving Time if it applies, then click Ok. You can change it later with the Change Time Zone button.

Step 4 — wait

The tool now does the work. Messages appear in the Excel status bar and on the Dashboard so you can see where it has got to:

Message	What is happening
Please wait, reading input file {1}/{2}	Opening your day files one by one.
Please wait, importing data...	Reading the merged data in.
Please wait, opening temp file...	Preparing the merge.
Please wait, writing temp file...	Merging them into a single dataset.
Please wait, importing data, {1}%	How far through the merge it is.
Please wait, calculating values...	Working out ΔT , energy and the derived columns.
Please wait, writing data to sheet...	Filling in the Data1 table.
Please wait, decoding error codes...	Translating any valve fault codes into plain English.
Please wait, building timeline charts...	Drawing the Dashboard charts.
Please wait, building the diagnostic Import sheet...	Rebuilding the raw Import sheet for diagnostics.
Please wait, applying number formats...	Formatting that table.
Please wait, updating setpoint and threshold lists...	Refreshing the setpoint and threshold pick-lists.
Please wait, recalculating the workbook...	Refreshing every report.
Please wait, recalculating the workbook... ({1} of {2})	Refreshing every report, in three passes.

Message	What is happening
Please wait, finishing up...	Last steps before handing back to you.
Please wait, clearing imported data... ({1} of {2})	Emptying the workbook (Clear All Data only — section 11).
Please wait, sorting data by timestamp...	Putting all samples in chronological order.
Please wait, building timeline chart {1} of {2}...	Drawing the Dashboard charts, one at a time.

A large import can sit on one message for many seconds. That is normal — let it finish.

Step 5 — the Import Details form

When the data is in, this form appears. Everything you type here feeds the reports.

Field	What to enter
Customer Name	The customer or building. Appears on every report and in the PDF file name.
AHU or HX Name	The equipment this valve serves, e.g. AHU-3 Chilled Water Coil.
Number of Parallel Valves	Leave at 1 unless several identical valves serve one coil.
Use valve-calculated design values	Tick this if you do not have the design figures. The tool derives them from the valve's own V'nom, P'nom and ΔT setpoint, and fills in the three fields below for you.
Design DeltaT	The design temperature difference across the coil, from the schedule.
Design Flow Rate	The design flow, from the schedule.
Design Power	The design capacity. Leave blank to have it calculated from flow and ΔT .
DeltaT V'min Flow Rate %	The flow, as a percentage of maximum, below which ΔT readings are ignored as unreliable. 25 is the usual starting point.
Electricity Cost (\$/kWh)	Your customer's electricity rate.
Pump Efficiency (%)	Wire-to-water efficiency of the pump.
Pump Head (feet)	Pump head.

The last three drive every dollar figure in the Savings report. If you do not know them, the defaults are reasonable industry averages — but say so when you present the numbers.

Click OK. The tool applies your entries, refreshes the reports, and returns you to the Dashboard.

Step 6 — the sensor mapping check

Immediately after the import, the tool cross-checks the valve's configuration against its own logged energy. If they disagree, you will see a message headed Sensor mapping check:

CHECK CONFIG — Cooling, but supply (T2) 57.9°F is WARMER than return 42.7°F — verify Installation Position.

What it means: the valve's energy registers show the coil was cooling, so the supply water must be the colder of the two sensors — but according to the valve's configured Installation Position, the sensor the tool has labelled 'supply' is the warmer one. Supply and return are therefore reported the wrong way round on every report.

What to do: check two things on site — the valve's Installation Position setting (is the meter really in the pipe it says it is?), and which pipe the remote sensor is strapped to. One of them is wrong. The tool reports the configuration faithfully; it cannot tell which of the two mistakes was made.

The same result appears in two other places: a red CHECK CONFIG flag beside Heating or Cooling on the Dashboard, and the Sensor Mapping row of the Performance report.

If the check passes, or if it cannot tell (no energy logged, or a changeover system running both heating and cooling), you see nothing at all. It only interrupts you when something is genuinely wrong.

Click SAVINGS in the banner at the top of any sheet. This is the customer-facing deliverable. Three buttons at the top right switch between three views of the same data.

View	Shows
Water	Water saved by the Energy Valve, in gallons, with a comparison bar chart.
Energy	Energy and dollars saved, plus an annualised projection.
Performance	How the valve is actually behaving — works whether or not the ΔT Manager was ever switched on. Use this one for commissioning and troubleshooting.



Figure 2 — the Performance view. Measured values on the left, automatic diagnostics on the right, and a plain-English summary underneath.

The measured values

Row	What it means
Avg Flow Rate	Average flow across the whole logged period.
Total Runtime	Hours the valve was actually passing water.
Total Volume	Total water moved.

Row	What it means
Avg Supply Temp	Average temperature of the water entering the coil. The sensor used (T1 or T2) is named in brackets — which one it depends on the valve generation and where the meter is installed.
Avg Return Temp	Average temperature leaving the coil.
Avg ΔT (Actual)	The temperature difference the coil actually achieved.
Design ΔT (Setpoint)	What it was designed to achieve.
Actual vs Design ΔT	The first number divided by the second. Below about 70% is worth investigating.
ΔT Manager Active %	How much of the period the ΔT Manager was doing its job. 0% means the feature is available but switched off.
Avg Power	Average thermal output.

The automatic diagnostics

The right-hand panel runs these checks and writes a plain-English verdict for each.

Check	What it is telling you
Actual vs Projected Flow	Whether the valve is delivering the flow the DDC asked for.
Supply Temp Stability	Whether the supply water temperature is holding steady. A supply that wanders by more than a few degrees can push the valve in and out of ΔT limiting on its own.
Flow Stability	Whether flow is hunting at a steady setpoint.
DDC Signal Status	Whether the control signal is modulating normally or thrashing.
Design Data	Whether the design figures came from the customer or were defaulted by the tool. Defaulted values are flagged with a red asterisk.
Flow vs ΔT Threshold	How much of the period ran above the V'min flow threshold, i.e. how much of the ΔT data is trustworthy.
Actuator Dithering	Whether the actuator is being commanded back and forth pointlessly, which wears it out. Reported as full-stroke equivalents.
Sensor Mapping	The physical cross-check described in section 5. Reads OK, CHECK CONFIG, or not checked.

Underneath the tables, a short paragraph summarises what the numbers mean in words you can read out to a customer.

Producing the deliverable

The envelope icon on the Savings sheet creates a PDF of the report and attaches it to a new e-mail. The file is named after the customer and equipment automatically. The printer icon on the Dashboard prints the current sheet.

7. Data Integrity

Click Data Integrity on the Dashboard, then Run Integrity Check. The tool scans the imported data and reports anything that would undermine the analysis. A summary table gives each section a PASS or an issue count; the detail table underneath lists every flagged item, with the exact Data1 cell address so you can go and look at it.

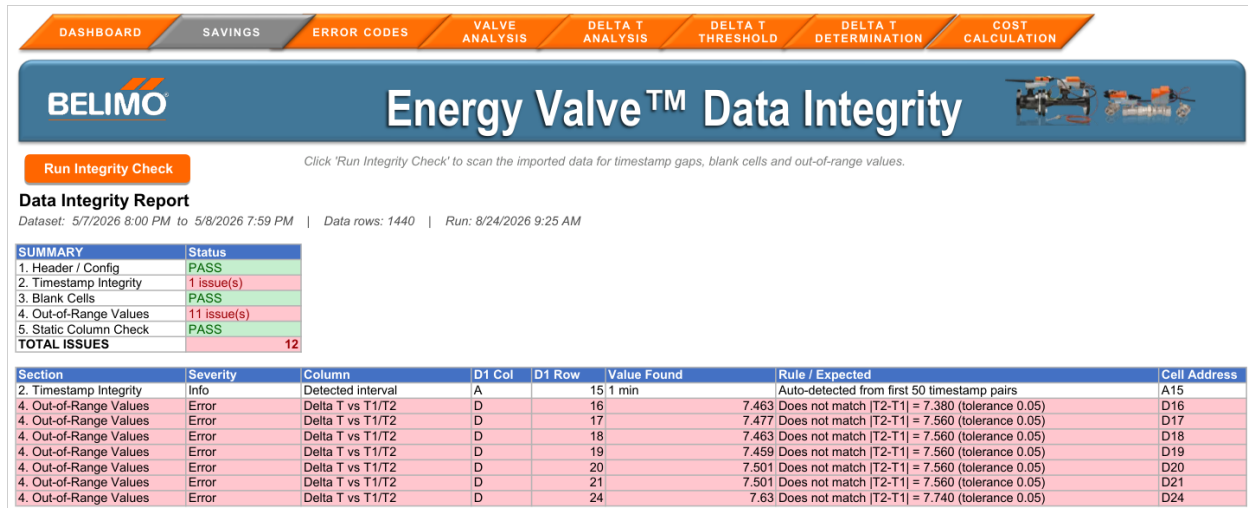


Figure 3 — a completed integrity check. Green is a pass, pink is something to look at. Here the ΔT column disagrees slightly with $T2$ minus $T1$ on eleven samples.

Section	What it checks
1. Header / Config	That the file's column layout is the one the tool expects.
2. Timestamp Integrity	Gaps, duplicates and out-of-order samples, and the logging interval it detected.
3. Blank Cells	Columns with holes in them.
4. Out-of-Range Values	Readings that are physically implausible, and consistency checks such as whether the reported ΔT matches $T2$ minus $T1$.
5. Static Column Check	Columns that never change, which usually means a sensor is not wired or not configured.

Run it whenever a report looks wrong, before you go looking for a fault in the plant — quite often the problem is the data, not the valve.

8. Advanced Mode

Click the spanner icon at the top right of the Dashboard, then Enter Advanced Mode. Extra sheets appear, reachable from the banner across the top of every screen. Enter Basic Mode puts them away again.

Sheet	What it is for
Valve Analysis	Detailed statistics broken down by control mode and by whether the ΔT Manager was active.
Delta T Analysis	Minutes spent in each control mode, with ΔT Manager status.
Delta T Threshold	Explores what would happen at different V' max settings.

Sheet	What it is for
Delta T Determination	The power-and- ΔT -versus-flow curve, with good and bad sample comparisons, used to establish an achievable ΔT .
Cost Calculation	The full cost workings behind the Savings report, including valve and install cost and payback period.
Data Integrity	As section 7.

Every one of these sheets repeats the same GENERAL INFORMATION block at the top — customer, equipment, valve size, whether zero-flow samples are included, the valve settings and the time frame — so you always know which data set you are looking at. The rest of this section describes each in turn.

Valve Analysis

The most detailed sheet in the tool. Everything is broken down three ways — Position Control, Flow Control and Power Control — and reported twice: once for the whole period, and once for only the time the ΔT Manager was active. Comparing the two columns is how you show what the ΔT Manager is worth.

- **Control Mode and Override Mode** — how long the valve spent in each control mode, and whether anything was overriding it rather than letting it modulate
- **Flow Set Point % and Relative Valve Position %** — the commanded and actual position, banded so you can see where the valve really lives
- **Flow in Gallons / Average Flow in GPM by valve position** — volume and average flow in each position band — this is where oversizing shows up
- **Low/Over Flow Calculator** — actual flow against the flow the DDC signal projected, with the difference. A persistent gap means the valve is not delivering what the controller is asking for
- **BTU/Energy Meter panel** — the meter's own display properties, for cross-checking against what the BAS shows
- **Daily / Weekly / Monthly** — the same figures projected out, for putting a period of logging into annual terms

Delta T Analysis

Answers one question: how much of the time was the ΔT Manager actually limiting, and in which control mode?

Pick a ΔT limiting value from the selector, and the sheet reports the minutes spent in Position, Flow and Power control with the ΔT Manager active, alongside counters for time spent in no control mode at all. Use it when you need to show that a valve was nominally capable of ΔT management but spent the period doing nothing of the sort.

Delta T Threshold

A what-if calculator for V'max. The upper block shows, for the V'max the valve is set to now, how much of the period ran above and below the ΔT threshold — count, average flow, average ΔT , total and percentage — both overall and while the ΔT Manager was active.

The lower block, Enter New V'max Value Below, recomputes all of it for a V'max you type in. That is the sheet's real purpose: before you recommend turning V'max down, use this to show the customer what it would have done to their operation over the period you logged.

Delta T Determination

Where you establish a ΔT setpoint the coil can actually hold, rather than the one on the schedule. The POWER AND DT VS FLOW chart plots power and ΔT against flow for the logged period; the saturation point — where more flow stops buying more power — is the ΔT the coil can really deliver.

- **DATA QUALITY OK?** — an indicator to check first. If the logged period does not contain enough spread of operation, the curve cannot be trusted and no calculation should be relied on
- **GOOD SAMPLE / BAD SAMPLE** — worked examples of what a usable curve and an unusable one look like, for comparison against yours
- **CALCULATE — STANDARD MODE** — derives the setpoint from the logged curve
- **CALCULATE — DESIGN** — works from the design figures you entered instead
- **CALCULATE — SATURATION POINT** — finds the point where added flow stops adding power. If the period never reached saturation the sheet says No Saturation Point Available — that is a valid answer, and means you need a period that includes high-load operation

Max. Flow (comfort) is calculated as 95% of the saturation flow — a deliberate margin so the recommendation does not sit right on the cliff edge.

Cost Calculation

The full workings behind the dollar figures on the Savings report. Customer Input at the top holds the three values you set with Update Energy Costs; everything below is derived.

- **% of Over Flow, per control mode** — how much more water was moved than the DDC signal actually called for
- **Total Water Pump Costs** — hours and cost in each control mode, split into ΔT Manager in standby versus active
- **Projected vs Actual Pump Costs** — what it would have cost against what it did cost
- **Waste Zone Calculator** — the cost attributable to running in the low- ΔT waste zone
- **Savings Estimator / Savings per Time Period / Annual Savings** — the same saving expressed over the logged period and annualised
- **Valve Cost, Install Cost, Total Cost, Pay Back in Years** — enter the installed cost of the valve and the sheet returns a payback period. These two cost fields are the only inputs on this sheet

Data Integrity

Covered in section 7. It is reachable both from Advanced Mode and from its own button on the Dashboard.

9. Dialog reference

Dialog	When it appears	What it asks
Unit Selection	During every import, and from Change Units	Temperature, flow, power and energy units; whether to remember them
Time zone	On the first import, and from Change Time Zone	Your time zone and whether daylight saving applies
Import Details	At the end of every import, and from Update Design Data	Customer, equipment, parallel valves, design figures, ΔT V'min, and the three energy cost inputs
Update Energy Costs	From the Update Energy Costs button	Electricity Cost (\$/kWh) · Pump Efficiency (%) · Pump Head (feet)

Dialog	When it appears	What it asks
Advanced User Mode	From the spanner icon	Enter Advanced Mode, Enter Basic Mode, or Exit
Sensor mapping check	After an import, only if the check fails	Nothing — it is a warning. See section 5, step 6.
About	From the blue i icon	Nothing — version and contact details

10. Settings you can change

Energy cost inputs

Set from the Update Energy Costs button, or on the Import Details form. These three values drive every dollar figure in the Savings report.

Setting	Default	Notes
Electricity Cost (\$/kWh)	0.12	Drives every dollar figure in both Savings views.
Pump Efficiency (%)	70%	Wire-to-water. Used for the over-pumping estimate.
Head in Feet (ft-head)	40	Used for the over-pumping estimate.

Everything else the tool uses — the thresholds behind the Actuator Dithering check, the CSV column map, the solver settings — is set centrally and is not adjustable from the screens described in this guide. If a check is reporting too many or too few events for a particular site, raise it with your Belimo contact rather than changing it locally.

11. Housekeeping

Clearing data

Normally you do not need to. The file cannot be saved, so closing it discards everything and the next person always starts from a clean template.

If you do need a guaranteed-empty copy to pass on — for instance after working on it with administrative access — ask your Belimo contact. There is an administrative reset that empties every imported row, the raw import, the customer details and all the reports in one step.

Versions

The version is shown at the bottom right of the Dashboard, under the button bar. This guide describes version 10.0.20. Always take the newest file from the shared folder rather than keeping a local copy — fixes go in regularly.

12. Troubleshooting

Symptom	What to do
Buttons do nothing	Macros are disabled. Close, unblock the file in its Properties, re-open and click Enable Content.
Excel says it cannot save	That is by design. See section 2. Print or e-mail the PDF instead.

Symptom	What to do
The tool looks frozen mid-import	It is almost certainly still working. Check the status bar at the bottom of the Excel window for the current step, and let it finish.
Import stops with “Not possible to delete the old temporary inputfile”	An old copy of the tool (from before August 2026) is being used on a PC that has also run a newer one. Use the current version from the shared folder and delete the old copy — one import with version 10.0.16 or later also clears the cause on that PC. If it persists, delete ~import.csv from %APPDATA%\Belimo (turn on Hidden items in File Explorer’s View menu to see it).
Supply and return look swapped	Look for the Sensor Mapping row on the Performance report and the CHECK CONFIG flag beside Heating or Cooling on the Dashboard. See section 5, step 6.
ΔT Manager Active shows 0%	The valve has the feature but it was never switched on for this period. That is a finding, not a fault — it is usually the recommendation you are there to make.
Savings figures are all zero	The Water and Energy views compare against ΔT -Manager-active operation. With the manager never active there is nothing to compare. Use the Performance view instead.
Numbers look implausible	Run Data Integrity before investigating the plant. Also check the design figures with Update Design Data — a wrong design ΔT skews everything downstream.
Wrong units	Change Units, then re-import. Unit changes apply to the next import, not retrospectively.